

MEMS Vibration Sensors for Smart Manufacturing

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ABSTRACT

This talk presents an in-house developed piezoelectric-MEMS vibration sensor module and its usage in smart manufacturing where Case (i) surface roughness prediction of fused silica substrate in a CNC polishing machine, Case (ii) fault diagnosis of a rolling bearing in an electric motor through their vibration signatures and machine learning (ML), and Case (iii) lifetime prediction of a drilling system are implemented as successful embodiments. In such a vibration sensing module, both PZT and AlN thin-film piezoelectric materials were verified while the differential configuration of the MEMS vibration sensor offers enhanced acceleration sensitivity for out-of-plane motion with common-mode noise rejection capability, therefore leading to high signal to noise ratio. The Thin-film Piezo on Silicon (TPoS) vibration sensor with a piezoelectric layer was fabricated using photolithography, frontal etching, and backside release. A fully integrated module comprising MEMS accelerometer, analog front-end, and a microcontroller was designed and characterized in a shaker system. The designed module provides a sensitivity of >100 mV/g, a resolution of $<1\text{mg}/\sqrt{\text{Hz}}$, and a bandwidth of 5kHz. The developed module features excellent compatibility with industrial 4.0 intelligent manufacturing applications.

BIOGRAPHY

Sheng-Shian Li (Senior Member, IEEE) received the B.S. and M.S. degrees in mechanical engineering from National Taiwan University, Taipei, Taiwan, in 1996 and 1998, respectively, and the M.S. and Ph.D. degrees in electrical engineering and computer science from the University of Michigan, Ann Arbor, MI, USA, in 2004 and 2007, respectively. In 2008, he joined the Institute of NanoEngineering and MicroSystems (iNEMS), National Tsing Hua University (NTHU), Hsinchu, Taiwan, where he is currently an NTHU Chair Professor. His research interests include microelectromechanical systems, integrated resonators and sensors, RF MEMS, CMOS-MEMS technology, frontend communication architectures, and integrated circuit design and technology. He received the Outstanding Research Awards from the National Science and Technology Council (NSTC) of Taiwan in 2020 and 2023 respectively. Together with his students, he received the Best Paper Awards at the 2011 IEEE International Frequency Control Symposium, the 2012 IEEE Sensors Conference, the 2017/2023 Transducers Conferences, and 2025 IEEE MEMS Conference. He



has served as the TPC/ETPC for the IEEE International Frequency Control Symposium, IEEE Sensors Conference, Transducers Conference, IEEE MEMS Conference, and IEEE IEDM. He also serves as the Associate Editors for the Journal of Microelectromechanical Systems, IEEE Sensors Journal, IEEE Sensors Letters, and IEEE Journal of the Electron Devices Society. He served as General Co-Chair for 2025 IEEE MEMS Conference.